DATA ANALYTICS PORTFOLIO PROJECT

Customer Shopping Behavior Analysis

Turning 3,900 Transactions into Revenue, Retention, and Growth Strategy

- Python · PostgreSQL · SQL · Power BI

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Business Intelligence & SQL Analytics Case Study

THE CHALLENGE

Business Problem

Shifting Customer Behavior, Unclear Drivers

A leading retail company is seeing measurable shifts in how customers shop across demographics, product categories, and channels — but leadership lacks a clear, data-backed view of what is actually driving those changes.

Management needs to know which factors — discounts, reviews, seasonality, or payment preference — truly influence purchase decisions and repeat buying, so investment in marketing and retention is not spent on guesswork.

Guiding Question: *"How can the company leverage shopping data to identify trends, improve engagement, and optimize marketing and product strategy?"*

WHY THIS MATTERS

$233K

in transaction revenue analyzed across the full customer base

27%

of customers are subscribers — a large untapped growth lever

80%

of customers already fall into the Returning or Loyal segment

SCOPE & INTENT

Project Objectives

Uncover Revenue Drivers

Quantify how gender, category, season, and shipping type shape spend and revenue mix.

Understand Customer Segments

Classify customers into New, Returning, and Loyal tiers to guide retention strategy.

Evaluate Discount Effectiveness

Identify which products depend on discounts and whether promotions grow high-value spend.

Deliver a Decision-Ready Dashboard

Translate SQL findings into an interactive Power BI view stakeholders can act on.

FOUNDATIONS

Dataset Overview

3,900
Transactions

18
Data Features

37
Missing Values (Review Rating)

100%
Cleaned & Validated

Customer Demographics

- Age & Age Group
- Gender
- Location
- Subscription Status

Purchase Details

- Item & Category
- Purchase Amount
- Season, Size, Color

Shopping Behavior

- Discount Applied
- Previous Purchases
- Review Rating, Shipping

PREPARATION PIPELINE

Data Cleaning & Feature Engineering

Raw CSV → Python (pandas) → Data Cleaning → Missing Value Handling → Feature Engineering → PostgreSQL

KEY TRANSFORMATIONS

- Age Group Creation — binned raw ages into four analysis-ready cohorts
- Purchase Frequency (Days) — engineered from raw purchase-cadence text
- Removed Redundant Columns — dropped promo_code_used (duplicate of discount flag)
- Missing Value Imputation — filled review ratings using category-level median

STRUCTURED QUERY ANALYSIS

SQL Business Analysis

All business questions below were answered using PostgreSQL — organizing raw transactions into queryable structure and powering every insight behind the dashboard.

Revenue by Gender

Compares total revenue generated by male vs. female customers

High-Value Discount Users

Finds discount shoppers who still spend above the average order

Top-Rated Products

Surfaces the 5 highest rated items by average review score

Shipping Analysis

Compares average spend across Standard vs. Express shipping

Subscriber Economics

Benchmarks avg spend & total revenue: subscribers vs. non-subscribers

Discount Dependency

Ranks products with the highest share of discounted purchases

Customer Segmentation

Buckets customers into New, Returning, and Loyal tiers

Category Leaders

Identifies the top 3 best-selling products within each category

Repeat-Buyer Loyalty

Tests whether repeat buyers (5+ orders) are more likely to subscribe

Revenue by Age Group

Quantifies revenue contribution across each customer age cohort

INTERACTIVE VISUALIZATION

Power BI Dashboard

What the Dashboard Shows

3.9K	Customers tracked with full segmentation filters
$59.76	Average purchase amount per transaction
3.75	Average review rating across all products
89.1	Average purchase-frequency in days

Filters let stakeholders slice results live by subscription status, gender, category, and shipping type — while linked visuals surface revenue and sales by category and by age group.

WHAT THE DATA REVEALS

Key Insights

Male shoppers drive 68% of revenue

Men generated $157,890 vs. $75,191 from women — a clear gender skew worth targeted campaigns.

80% of customers are already loyal

3,116 of 3,900 customers fall in the Loyal segment; only 83 are brand-new, signalling strong repeat behavior.

Non-subscribers generate 73% of revenue

Non-subscribers total $170K vs. $63K from subscribers — subscription upside remains largely untapped.

Hats & Sneakers over-rely on discounts

50% and 49.7% of their purchases involve a discount, the highest dependency of any product line.

Express shipping outspends Standard

Express orders average $60.48 vs. $58.46 for Standard — a small but consistent premium-service lift.

Young Adults lead age-group revenue

Young Adults contribute $62,143, narrowly ahead of Middle-aged, Adult, and Senior cohorts.

STRATEGIC DIRECTION

Business Recommendations

Promote Subscriptions

Bundle exclusive perks and early access to convert the 73%-revenue non-subscriber base.

Build Loyalty Programs

Reward the 701 Returning customers with tiered incentives to graduate them into Loyal.

Rebalance Discount Policy

Cap discount depth on Hats & Sneakers to protect margin without losing volume.

Elevate Top-Rated Products

Feature Gloves, Sandals, Boots, Hat, and Skirt in campaigns to lift conversion.

Target High-Value Segments

Prioritize male shoppers and Young Adult / Express-shipping customers in media spend.

Personalize by Category

Use category-level revenue mix (Clothing-led) to guide merchandising and inventory.

END-TO-END PROCESS

Project Workflow

1
Dataset
3,900 raw transaction records

→

2
Python
Cleaning & feature engineering

→

3
PostgreSQL
Structured relational storage

→

4
SQL Analysis
17 business questions answered

→

5
Power BI
Interactive stakeholder dashboard

→

6
Business Insights
Strategy & decision support

WRAPPING UP

Conclusion

This project turned 3,900 raw transactions into a governed, queryable dataset and a decision-ready Power BI dashboard — revealing exactly which customers, categories, and channels drive revenue and loyalty for the business.

The findings show that data-driven decisions — in discount policy, subscription strategy, and customer segmentation — can materially improve both customer satisfaction and revenue outcomes, turning analytics into a repeatable growth lever for the business.

End-to-End

Python → SQL → BI pipeline

17

Business questions answered

6

Strategic recommendations

Thank You

Let's discuss how data can drive your next business decision.

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